CS401: Advanced Computer Science & AI Systems

Lecture Notes & Exam Study Guide — Department of Computer Science

---

Module 1: Operating Systems & Concurrency

Operating systems act as an intermediary between user applications and bare computer hardware. A process is a program in execution containing a program counter, stack, data section, and heap.

1.1 Deadlock Characterization (Coffman Conditions)

A deadlock occurs when a set of processes are blocked because each process is holding a resource and waiting for another resource held by some other process. A deadlock situation can arise if and only if all four Coffman conditions hold simultaneously:

1. **Mutual Exclusion:** At least one resource must be held in a non-shareable mode.
2. **Hold and Wait:** A process must be currently holding at least one resource and requesting additional resources.
3. **No Preemption:** Resources cannot be forcibly preempted from a process holding them.
4. **Circular Wait:** A closed loop of processes exists, where each process holds at least one resource needed by the next.

1.2 Virtual Memory & Page Replacement

Virtual memory separates user logical memory from physical memory. The Memory Management Unit (MMU) translates virtual addresses into physical addresses using page tables. When a requested page is not resident in physical RAM, the OS triggers a **page fault interrupt**. Common page replacement algorithms include FIFO (First-In-First-Out), LRU (Least Recently Used), and Optimal (Belady's Algorithm). Belady's Anomaly demonstrates that for FIFO, increasing the number of page frames can counterintuitively increase page faults.

Module 2: Algorithmic Complexity & Graph Search

Asymptotic analysis evaluates algorithm efficiency as input size n approaches infinity. Big-O represents the asymptotic upper bound: f(n) = O(g(n)) if there exist positive constants c and n0 such that 0 <= f(n) <= c*g(n) for all n >= n0.

Algorithm	Data Structure	Average Time	Worst Case Time	Space Complexity
Quicksort	Array	O(n log n)	O(n^2)	O(log n)
Mergesort	Array / Linked List	O(n log n)	O(n log n)	O(n)
Dijkstra's	Adjacency List + Min-Heap	O((V + E) log V)	O((V + E) log V)	O(V)
Bellman-Ford	Edge List	O(V * E)	O(V * E)	O(V)
Binary Search	Sorted Array	O(log n)	O(log n)	O(1)

2.1 Dijkstra's Shortest Path Algorithm

Dijkstra's algorithm solves the single-source shortest path problem for directed or undirected graphs with non-negative edge weights. It maintains a priority queue of unvisited vertices sorted by tentative distance from the source vertex. Crucial Note: Dijkstra's fails when graphs contain **negative edge weights**; for graphs with negative weights, the **Bellman-Ford algorithm** must be used instead.

Module 3: Artificial Intelligence & Machine Learning

Machine Learning focuses on algorithms that learn patterns from empirical data rather than relying strictly on rule-based programming.

3.1 Supervised vs Unsupervised Learning

Supervised Learning: The training dataset contains both feature inputs X and ground-truth labels Y. Tasks are categorized into Regression (predicting continuous real-valued outputs) and Classification (predicting discrete category labels). Examples include Linear Regression, Logistic Regression, Support Vector Machines (SVM), and Random Forests.

Unsupervised Learning: The dataset contains only feature vectors X with no target labels. The model must discover latent structure, groupings, or representations. Primary tasks include Clustering (K-Means, DBSCAN) and Dimensionality Reduction (Principal Component Analysis - PCA, t-SNE).

3.2 The Bias-Variance Tradeoff & Overfitting

The total expected generalization error of a machine learning model is composed of three components: **Error = Bias^2 + Variance + Irreducible Error**.

- **High Bias (Underfitting):** Occurs when the hypothesis class is too restrictive to capture underlying relationships. The model suffers high training error and high validation error.
- **High Variance (Overfitting):** Occurs when the model fits training noise and random fluctuations rather than the true signal. The model shows extremely low training error but high validation/test error.
- **Mitigation Strategies:** Regularization (L1 Lasso for feature sparsity, L2 Ridge for weight shrinkage), Cross-Validation (k-fold), Dropout in neural networks, and increasing training data volume.

Module 4: Database Normalization & ACID Properties

Database normalization organizes relational schema to reduce data redundancy and eliminate update, insertion, and deletion anomalies.

4.1 Normal Forms Hierarchy

1. **First Normal Form (1NF):** Each column contains atomic (indivisible) values, and each record is unique.
2. **Second Normal Form (2NF):** In 1NF and no non-prime attribute is partially dependent on any candidate key (eliminates partial dependency).
3. **Third Normal Form (3NF):** In 2NF and no non-prime attribute depends transitively on a candidate key (X -> Y where Y is non-prime implies X is superkey or Y is prime).
4. **Boyce-Codd Normal Form (BCNF):** For every functional dependency X -> Y, X must strictly be a superkey.

4.2 ACID Transaction Guarantees

- **Atomicity:** 'All or nothing' — transactions execute entirely or roll back cleanly.
- **Consistency:** Transitions the database strictly from one valid state to another satisfying all schema invariants.
- **Isolation:** Concurrent transactions execute without interfering with one another (serializability).
- **Durability:** Committed transactions persist permanently across power outages and crashes using write-ahead logs (WAL).

High-Yield Exam Review Summary

Key Formula 1 (Amdahl's Law): Speedup = 1 / ((1 - p) + (p / s)), where p is parallel proportion and s is speedup.
Key Formula 2 (F1 Score): F1 = 2 * (Precision * Recall) / (Precision + Recall).
Key Formula 3 (Cross-Entropy Loss): L = - sum(y_i * log(p_i)) for multi-class classification.